

## **Case Study 10: Corporate Financial Performance & Forecasting**

### **Scenario:**

A multinational company wants to analyse its financial statements, forecast revenue, and identify profitable product lines.

### **Advanced Excel (10 tasks)**

- Build P&L, Balance Sheet, Cash Flow models.
- Pivot tables for revenue by product.
- INDEX-MATCH for dynamic lookups.
- Waterfall chart for profit breakdown.
- Forecast sales using exponential smoothing.
- Scenario analysis for cost changes.
- Automate quarterly reports with macros.
- Dashboard for financial KPIs.
- Link Excel outputs to SQL queries.
- Export data for Python analysis.

### **SQL (10 tasks)**

- Create tables for sales, costs, profits.
- Query quarterly revenue growth.
- JOIN product and region tables.
- GROUP BY profitability by segment.
- Stored procedures for financial KPIs.
- Window functions for YoY growth.
- Query top 5 profitable products.
- Trigger alerts for cost overruns.
- Export SQL results to Excel.
- Link SQL outputs to Python.

### **Python (10 tasks)**

- Import SQL financial data into Pandas.
- Clean and preprocess revenue data.
- EDA on profitability trends.
- Regression models for revenue forecasting.
- ARIMA for time-series forecasting.

- Visualize profit margins with seaborn.
- Automate quarterly reporting.
- ML model for cost prediction.
- Export results to CSV for Power BI.
- Link Python outputs to Data Science.

#### **Data Science (10 tasks)**

- Feature engineering: profit margin drivers.
- Clustering for product segmentation.
- Classification models for high vs. low profit.
- Random Forest for cost prediction.
- PCA for dimensionality reduction.
- Cross-validation for forecasting models.
- Compare regression vs. ARIMA.
- Confusion matrix for classification.
- Predictive insights for CFO.
- Export results to Power BI.

#### **Power BI (10 tasks)**

- Import datasets from Excel, SQL, Python.
- Dashboard: revenue, cost, profit trends.
- Slicers for product categories.
- YoY growth visualization.
- Drill-through for regional analysis.
- KPI cards for profitability.
- Forecasting visuals for revenue.
- Scenario-based dashboards.
- Publish dashboard for executives.
- Final integrated dashboard: *Corporate Financial Performance*.

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These case studies simulate **real-world finance problems**:

- **Case Study 1** → Banking risk management.
- **Case Study 2** → Corporate profitability forecasting.

***Corporate Financial Performance & Forecasting*** — demonstrate how **Advanced Excel, SQL, Python, Data Science, and Power BI** can be interlinked to solve real-world financial challenges. Each section builds progressively: Excel for structured modelling, SQL for data management, Python for automation and predictive analytics, Data Science for advanced modelling, and Power BI for visualization. By culminating in an integrated dashboard, learners not only practice technical skills but also deliver actionable insights for decision-makers, bridging academic learning with industry application